Prop-Firm Challenges: A Barrier Model of Pass Rates and Expected Value

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Abstract

Proprietary-trading "evaluation challenges" — paid tests that, once passed, grant a simulated "funded" account paying a share of simulated profits — have become a large retail product, yet their economics have not been analyzed rigorously. We model a challenge as a barrier-crossing problem. For a driftless participant facing a fixed loss limit, optional stopping gives a pass probability of $L/(T+L)$, the ratio of the loss limit to the sum of loss limit and profit target, independent of volatility. We show that this is a ceiling rather than a value: the loss limit trails the account's high-water mark, positions are flattened daily, and frictions are charged per trade, so the attainable rate falls below it. On six years of one-minute index-futures data we decompose the gap — 44.4% under the bare geometry, 40.4% once the daily bracket and its truncation are imposed, 34.7% once the floor trails — and we show the ceiling is unattainable from either direction, since larger positions break the continuity the theorem assumes while smaller ones pay a fixed cost per trade. We then price the full life cycle — evaluation, funded lock, qualification and withdrawals — on realized paths. Expected value per evaluation is $+0.052\,L$ at the fee actually paid and $-0.018\,L$ at list price. The consistency rule that operators present as a hurdle is met with exact equality by a repeated fixed bracket and costs nothing, but the design sits precisely on that boundary. We derive the population pass rate at which the seller breaks even — 20% at the fee paid, 40% at list price, against 34.7% for the mechanical participant we model — and show the two are consistent. The economics assume no directional edge and are therefore a lower bound for the buyer.

Keywords: prop firm challenge; funded trader; barrier crossing; optional stopping; expected value; consumer protection.
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Declaration of generative AI use. During the preparation of this manuscript the author used Anthropic's Claude (via the Claude Code interface) to assist with drafting and structuring the exposition, formalizing notation and proofs, and writing and checking analysis code. All research design, data, backtests, modeling choices, results, interpretation, and conclusions are the author's own; the author reviewed and verified all content and takes full responsibility for it.

1 Introduction

Over the last few years a retail financial product has grown from nothing to substantial scale: the proprietary-trading evaluation challenge. A customer pays a fee to attempt a trading test on a simulated account; on reaching a profit target without breaching a loss limit, they receive a "funded" account and a share of the profits they generate on it. The accounts are simulated and the provider is the counterparty, so a payout is a marketing and retention expense of the firm rather than a share of market proceeds.

Supervisory authorities have issued consumer warnings about these products. Consob (8 July 2024) describes them as "a kind of finance video game" and records complaints both about

"the level of difficulty of the tests, which are allegedly contrived to push 'players' to try again" and about "the failure to share the alleged profits"; the FSMA (7 March 2024) notes that the firms "do not hold any authorization" and that consumers "often have to take, and pay for, several of them"; the CNMV has issued an equivalent warning. Industry sources publish aggregate pass rates. But we are not aware of a formal model of the pass probability, or of an expected-value estimate for the buyer, and that is what this paper supplies.

The paper makes three contributions.

1. *A closed-form ceiling on the pass probability.* For a driftless participant facing a fixed loss limit, the probability of passing is $L/(T+L)$, independent of volatility (Section 3). We show that the product's actual design departs from every assumption the identity requires, so the identity bounds the pass rate rather than giving it, and we measure the gap.

2. *A characterization of the design levers.* The trailing locking drawdown, the per-payout cap and the multi-cycle thinning of the cushion move the buyer's expected value below the bare geometry, while the moving consistency target is met with equality by a repeated fixed bracket. We quantify each on realized paths.

3. *An empirical pricing of the full cycle, and of the seller's position.* We estimate expected value per evaluation at the fees actually observed and at list price, and derive the population pass rate at which the operator breaks even.

2 Institutional background

A standard challenge has two stages. In the **evaluation**, the customer must grow a simulated balance by a profit target T before its drawdown reaches a loss limit L; the attempt costs a fee f. Once passed, the account is **funded**: the customer keeps trading and receives, at each qualifying withdrawal, a share s of $\min(\text{cycle profit}, \kappa)$, up to N withdrawals. **Throughout we take the loss limit L as the unit of account and report every monetary quantity as a multiple of it**; for the size class we estimate on, L is of the order of one thousand dollars.

Two features stand out. The loss limit and its locking threshold are set out in the funded-account agreement we hold and are confirmed on a live dashboard. The profit target and the consistency cap are *not* in that agreement — they are rules of the evaluation stage, which is governed by a separate document we do not have — and are known to us only from a live dashboard, on two accounts. We treat them as parameters of a standard family and flag the resulting uncertainty where it bites.

- **The loss limit is a hard, intraday barrier.** The floor is breached the moment floating equity touches it, not at the close, so the cushion (balance minus floor) is a broker-imposed stop. Before locking, the floor trails the running equity high on closing balances, so gains raise the floor rather than widening the cushion, and it locks permanently once net profit exceeds a threshold Λ.[1]

- **A moving ("consistency") target.** No single day may contribute more than a share c of total profit ($c = 1/2$), which is equivalent to a target that rises with the best day: $T_{\text{eff}} = \max(T_{\text{base}}, m/c)$ where m is the largest single green day. On the dashboard, a best day of $0.6445\,L$ ($644.50) displayed a target of $1.289\,L$ ($1,289.00) rather than the 1.25 base. The funded stage has no such rule.

[1] The agreement's wording — the loss limit becomes the initial balance plus $0.1\,L$ "once the account is above the initial trail balance" — is ambiguous between $26\,L$ and $26.1\,L$; we use $26.1\,L$, the value shown on the live dashboard, which is the conservative reading. With the initial balance at $25\,L$ this gives $\Lambda = 1.1\,L$ and a locked floor of $25.1\,L$.

We summarize a challenge by $E = (f, L, T, c, \Lambda, \kappa, s, N)$. All of L, T, Λ, κ are quoted by the operator in fixed currency, a fact that matters in Section 3. The normalized family is given in Table 1; these are the estimated values of a standard family, not the terms of any one provider.

Table 1: The normalized parameter family.

Parameter	Symbol	Value
Profit target	$\theta = T/L$	1.25
Lock threshold	$\lambda = \Lambda/L$	1.10
Consistency cap	c	1/2
Entry fee	$\varphi = f/L$	0.05/0.07/0.10/0.14
Notional account size	A/L	25
Profit split	s	0.9
Payout	—	$s \cdot \min(0.5 \cdot \text{profit}, 1\,L)$, floor $0.5\,L$
Qualifying-day threshold	—	$0.10\,L$
Operating float	—	$2\,L$

2.1 Related work

The product has not, to our knowledge, been modelled in the academic literature, but it is not undocumented. Three European supervisors issued consumer warnings in 2024: Consob (8 July 2024) characterises the offering as "a kind of finance video game" and records complaints about the difficulty of the tests and about unpaid profit shares; the FSMA (7 March 2024) notes that the firms "do not hold any authorization and therefore are not allowed to provide investment services"; the CNMV has issued an equivalent warning, which we know from the Consob release rather than from the primary document. The nearest empirical literature concerns the populations these products draw on rather than the products themselves: the survival of retail day traders (Barber, Lee, Liu and Odean, 2014; Chague, De-Losso and Giovannetti, 2020), and the regulatory record on leveraged retail products, where ESMA's 2018 product-intervention decision reports that 74 to 89% of retail CFD accounts lose money with average losses of €1,600 to €29,000. The structural analogy is to tournament incentives, where a convex payoff schedule induces risk-taking by the trailing participant (Brown, Harlow and Starks, 1996). On the question of whether retail signals carry exploitable information, Darmanin (2026), an unrefereed working paper, tests five signal families under multiplicity control and finds four refuted and the remainder inconclusive — the same distinction we draw below between an absent effect and an undetectable one. Demand for a break-even-to-negative product with a small chance of a salient payoff is consistent with the lottery-preference literature (Kumar, 2009; Barberis, 2012) and with cumulative prospect theory (Tversky and Kahneman, 1992).

3 A barrier model of the challenge

Setup. Let X_t be the participant's cumulative profit and loss, $X_0 = 0$. A *skill-free* participant is one whose X is a continuous martingale. The evaluation is passed at the first time X reaches $+T$ and failed at $-L$.

The pass probability is bounded by geometry.

Proposition 1 (geometric ceiling)**.** *Let X be a continuous martingale, and let the evaluation be passed at $+T$ and failed at $-L$, with L measured from a* fixed *floor. If $\tau = \inf\{t : X_t \notin (-L, T)\}$ is almost surely finite — non-degenerate volatility and an unbounded horizon — then optional stopping gives $P(\text{pass}) = L/(T + L)$, independent of the volatility of X. If the floor* trails *the*

running maximum, if X has jumps so that X_τ overshoots the barrier it crosses, or if the horizon is bounded, then

$$P(\textit{pass}) \;\le\; \frac{L}{T+L},$$

with equality only in the fixed-floor, continuous, unbounded-horizon case.

All three departures are present in the product: the loss limit is an end-of-day trailing drawdown, the participant trades a position size for which a single session's adverse excursion is several times the cushion, and positions are flattened at the close.

Proof. On $[0,\tau]$, X is a bounded martingale, so optional stopping gives $\mathbb{E}[X_\tau] = 0$. In the fixed-floor continuous case $X_\tau \in \{T, -L\}$ with probabilities p and $1-p$, whence $pT - (1-p)L = 0$ and $p = L/(T+L)$; volatility enters $\mathbb{E}[\tau]$ but not p. With jumps, X_τ overshoots, so $\mathbb{E}[X_\tau] = 0$ gives $p\,\mathbb{E}[X_\tau \mid \text{pass}] - (1-p)\,\mathbb{E}[-X_\tau \mid \text{fail}] = 0$ with $\mathbb{E}[X_\tau \mid \text{pass}] \ge T$ and $\mathbb{E}[-X_\tau \mid \text{fail}] \ge L$, and the inequality follows. With a trailing floor, write $\tau_{\text{tr}} = \inf\{t : M_t - X_t \ge L\}$ and $\tau_{\text{fx}} = \inf\{t : X_t \le -L\}$, where $M_t = \sup_{s\le t} X_s$. Since $M_t \ge X_0 = 0$, any t with $X_t = -L$ already satisfies $M_t - X_t \ge L$, so $\tau_{\text{tr}} \le \tau_{\text{fx}}$ on every path; hence $\{X \text{ reaches } T \text{ before } \tau_{\text{tr}}\} \subseteq \{X \text{ reaches } T \text{ before } \tau_{\text{fx}}\}$ and the trailing-floor probability is no larger. □

This is the classical gambler's-ruin identity (Feller, 1968; Karatzas and Shreve, 1991), equivalently the value of a double-barrier claim (Harrison, 1985); the novelty is the object to which it is applied and the demonstration that the object violates its hypotheses. With the account parameters of Section 2, $\theta = T/L = 1.25$, the ceiling is $1/(1+\theta) = 1/2.25 = 44.4\%$.

Frictions. Commissions and slippage impart a negative drift. Measured on the estimation window for the evaluation bracket, one round turn costs $0.0035\,L$ in commission and one tick of adverse fill on both entry and stop, and the mean profit and loss of a single day is $-0.0191\,L$, about 1.9% of the cushion per session. For a Brownian X with variance σ^2,

$$P(\text{pass}) = \frac{1 - e^{-2\mu L/\sigma^2}}{1 - e^{-2\mu(T+L)/\sigma^2}},$$

which reduces to $L/(T+L)$ as $\mu \to 0$ and lies below it for $\mu < 0$; at the measured drift this alone removes about 1.7 percentage points. Note also that the volatility independence of Proposition 1 is exact only for a fixed number of round turns: because commission is charged per trade rather than per unit of time, a strategy that trades more often to cover the same distance pays more, so trading frequency reintroduces an indirect dependence.

The consistency (moving-target) rule. The consistency cap makes the target dynamic: $T_{\text{eff}} = \max(T_{\text{base}}, m/c)$ where m is the largest single green-day profit and $c = 1/2$, so $T_{\text{eff}} = \max(T_{\text{base}}, 2m)$.

Proposition 2 (the consistency cap is exactly non-binding for a fixed bracket)**.** *The cap* $\max(d_1,d_2)/(d_1+d_2) \le 1/2$ *holds if and only if* $d_1 = d_2$*. A participant taking the same bracket on both green days nets the same amount twice — the profit target is taken with a limit order, which fills at the limit, and commissions are deterministic — so the cap is met* with equality *and the two-day pass carries no expected-value cost. This is a knife edge, not a margin. Any deviation from the repeated bracket forces a top-up day: sized to the exact shortfall it costs about* $0.005\,L$ *per evaluation; sized to a fixed* $0.1\,L$ *target, the size used in our own live template, it costs about* $0.016\,L$*.*

The result holds for a participant who meets three operational conditions, and fails otherwise. **One trade per day**: the cap compares the largest day against the total, so a day assembled

from several trades nets an arbitrary amount and the equality is lost — this is exactly what happened on the one live account we observe, where the first day was traded three times. **The identical target on both days**: 128 ticks, not "about $0.64\,L$"; the equality is exact or it does not exist. **Reading the shortfall from the operator's dashboard before requesting the pass**: the operator publishes the effective target and the remaining amount, and those are the arbiter, not the participant's own arithmetic.

No two-day pass has any margin. Because the effective target is exactly twice the largest day, two days days sum to it only if they are identical, so every two-day pass lands at precisely 50.000% with a remaining balance of exactly zero — whatever the target chosen. Sweeping the daily target confirms it. On a 704-session sub-window, with 564 contiguous overlapping accounts rather than the bootstrap used elsewhere, any design whose net daily profit exceeds half the base target passes in two days with 97% of those passes carrying less than $0.001\,L$ of margin over the effective target, while any design whose net daily profit stays below half the base target never raises its own target, is forced structurally into a third day, and carries such a margin on none of its passes — at the cost of a lower per-day win probability. That trade-off, and not the nominal 50% figure, is the economic content of the consistency rule.

Two features of the operator's implementation are not established by any document we hold. First, whether the cap is inclusive ($\leq 1/2$) or strict ($< 1/2$): secondary descriptions say the best day "cannot exceed" half of total profit, which reads as inclusive, but the evaluation-stage agreement is a separate document we do not have, and the funded-account agreement does not mention the rule. Second, whether the effective target displayed by the dashboard is the exact quantity $2m$ or a rounded version of it: of the two dashboard states we observe, one matches $2m$ to the cent ($0.6445\,L \to 1.289\,L$) and one does not ($0.8248\,L \to 1.6496\,L$ displayed as $1.65\,L$); in dollars, \$644.50 → \$1,289.00 and \$824.80 → \$1,649.60 displayed as \$1,650.00. Our base case assumes an inclusive cap and an unrounded target, under which a repeated fixed bracket clears the cap with equality at no cost. If either assumption fails, the two-day pass is unattainable, and what it costs depends on whether the participant has a cheap way over the line: with a top-up day sized to the shortfall, about $0.001\,L$ per evaluation; without one, the pass rate falls from 34.8% to 21.4% and expected value from $+0.053\,L$ to $+0.004\,L$ at the fee paid (Table 5). This is the single assumption on which Proposition 2 turns.

A further caveat applies to the fill mechanics. A resting limit order can in principle fill better than its limit if price trades through it within the bar. At one-minute resolution we cannot observe this — we find no instance in 990 target-hitting sessions, but the test cannot see inside the bar — so we report it as unmeasured rather than absent. Its asymmetry runs against the participant: a better fill makes that day the largest and raises the effective target. Separately, commissions are modelled as a constant $0.0035\,L$ per round turn, the value observed in the account history; exchange and regulatory fees can move it by amounts of the order of $0.00002\,L$, and that is enough to break the exact equality on which the two-day pass depends.

The trailing, locking drawdown (funded stage). Write B_t for the funded balance and $M_t = \sup_{s \leq t} B_s$ for its running maximum. While the floor trails, $F_t = M_t - L$, so the usable cushion is $C_t = B_t - F_t = L - DD_t$, where DD_t is the current drawdown from the peak. The account dies exactly when its drawdown from the peak reaches L, and locks when the balance first gains Λ. Reaching $+\Lambda$ before a drawdown of L is a first-passage problem on the running-maximum drawdown, path-dependent, with no elementary closed form (Magdon-Ismail et al., 2004). By the pathwise domination in the proof of Proposition 1, applied with target Λ and floor L, $P(\text{lock}) \leq L/(\Lambda + L)$.

The lock is optimally taken in a single day. Splitting it across more days re-exposes the account to the trailing barrier, whose floor ratchets up on each new high, and so lowers survival; and because commissions and slippage make the funded game subfair, bold play is optimal in the Dubins–Savage sense (Dubins and Savage, 1965). Modelled as one bracket whose target

sits just beyond the lock threshold — 240 ticks against a 192-tick floor — this gives $P = SL/(TP + SL) = 192/(240 + 192) = 44.4\%$.

The ceiling is unattainable from either direction. The ceiling is not attained from either direction, and the reason is instructive. A participant sizing large relative to the cushion violates the continuity the theorem assumes: with one full contract against a cushion of $1\,L$, the median adverse excursion of a single unbracketed session is $1.66\,L$ and two sessions in three would breach the floor outright — the first increment of the process is four times the distance to the barrier. A participant sizing small restores quasi-continuity but pays a fixed friction per trade: at $0.064\,L$ per step, commission plus one tick of slippage on entry and on the stop is 3.4% of the step, and roughly 180 steps are required. Sweeping the position size at a constant 2:3 target-to-stop ratio, the pass probability peaks at the largest single-contract bracket — 40.4% against a 44.4% ceiling under a fixed floor — and falls away in both directions: 28.4% at three micros, 0.5% at a single micro. The trailing floor then removes a further 5.7 points at that optimum, and proportionally more as the step shrinks: a sequence of small winning days raises the floor by the amount won and the cushion never widens, so high-win-rate, small-step strategies are exactly the configuration a trailing drawdown punishes most.

Expected value of the full product. The funded account is an absorbing renewal process: each qualifying cycle pays $s \cdot \min(0.5 \cdot \text{profit}, \kappa)$, up to N, with the cushion thinning at every withdrawal, since withdrawals leave the balance while the locked floor does not fall. Writing V for the expected total payout of a funded account,

$$EV_{\text{buyer}} = -f + P(\text{pass}) \cdot V,$$

whose sign is governed by the levers above.

4 Data and methodology

Data and instrument. One-minute bars of an E-mini index future over 1,713 real trading sessions spanning 2020–2026, a window that includes a sharp crash, a full bear market, and surrounding bull phases, at the price-volatility scale of the contract as it trades today. Barriers are evaluated on the realized intraday paths, with no assumed volatility or return distribution.

Intraday data are licensed from a commercial vendor under terms that do not permit redistribution. The simulation code is available from the author on reasonable request. Independent verification would require equivalent one-minute data for the same contract and window.

Skill-free baseline. At most one trade per session, entered thirty minutes after the regular session open, with fixed currency-denominated brackets matching each phase's rules, entered at market with the side chosen at random. Commissions are the confirmed values ($0.0035\,L$ round-trip per full contract; $0.0012\,L$ per micro, so $0.0036\,L$ for the three-micro qualification trade) and one tick of slippage is applied on entry and on the stop; the account is flattened at session end, so an unresolved day (1.9% of sessions at minute 0, rising to 8.0% by minute 180) earns only the mark-to-market at the close. The full life cycle is simulated by bootstrap over the real sessions.

Death convention. The daily stop is truncated to the cushion, $sl = \min(sl_{\text{nominal}}, \text{cushion} - 0.04\,L)$; the account dies when $sl < 0.04\,L$; and the race is recomputed with the truncated stop, so a truncated stop also lowers the probability of reaching the target. This convention is used for every figure in the paper.

The ceiling as a consistency check. The optional-stopping identity bounds the probability of reaching $+T$ before $-L$ over an unbounded horizon. The per-day statistic we report is a bounded-horizon quantity — positions are flattened at the session close — so the bound applies to it only when the unresolved-by-close rate is negligible. We therefore report that rate alongside each phase (2.6%, 5.4% and 2.1%) and treat the comparison as a consistency check rather than a proof of correctness. On windows where the contract traded at a much lower price, and the same currency bracket was therefore wide relative to the session range, the unresolved rate rises and the conditional win rate exceeds the nominal ceiling; the check is informative only in the regime we estimate on.

Internal cross-check. The cycle was implemented twice by the author, in Python on the full contract and in NinjaScript on the micro contract, over different data spans. The two agree on the qualitative funnel — both reproduce the $P \approx p^2$ identity — but differ materially on levels (35.6% versus 27.8% pass; $0.087\,L$ versus $0.051\,L$ per evaluation), and the second implementation was never re-run on the estimation window used here. This is an internal consistency check, not independent replication.

The directional baseline. The economics are computed for a participant with no directional edge. A session-level test of the entry levels against a matched random-side control, over 704 sessions with a block bootstrap over 15-session blocks, gives $-0.005R$ with a 95% interval of $[-0.026, +0.016]$, at a minimum detectable effect of $0.021R$. These directional tests run on a 704-session sub-window of the estimation period, to which the pre-registered grid is wired; porting it would require re-deriving the session features it depends on. The economics assume no directional edge rather than relying on the absence of one.

Portfolio and capital. We simulate K accounts against a common pool of real sessions and report results under three cross-account policies. Under **simultaneous entry with independent sides** — the policy our tables use — every live account takes a position in the same session at the same entry minute, each on an independently drawn side. Under a **sequential queue**, at most one position is open across the book at any instant and the next account enters at the minute the previous one's bracket closes; because the median bracket resolves in about fifteen minutes inside a 405-minute session, the queue is nearly non-binding up to about eight accounts and buys the same number of evaluations. Under **simultaneous entry on a shared side**, all accounts take the same direction. At $K = 5$ and the fee paid, the median and net-loss probability are $44.723\,L$ and 2.0% under the queue, $46.286\,L$ and 3.4% under simultaneous independent sides, and $2.82\,L$ and 39.7% under a shared side. The reported policy is therefore intermediate: more adverse than the queue, far less adverse than a shared direction. A shared directional bias across accounts destroys the result under all three: it removes the decorrelation on which every multi-account figure depends. We report the multi-account figures because they characterize the rule structure, not as an operating design: see the limitations on multi-account prohibitions in Section 7.

The modelled participant abandons an evaluation after 45 trading days. This is a parameter of the participant, not of the product: the trading rules we hold impose no time limit on an evaluation, only closure of a funded account after thirty consecutive calendar days without trades. The cap does not bind for the bracket we model — the pass rate is 40.4% under a fixed floor and 34.7% under the trailing floor with the cap at 45 days and unchanged at 600 — but it binds heavily for smaller brackets, which take 20 to 300 sessions to reach the target.

We price the product as a repeated purchase financed from a fixed working balance of $2\,L$, with each payout removed from the balance rather than reinvested. The float size is a modelling parameter chosen to make the front-loaded risk visible, not a recommendation.

Table 2: Skill-free win rate against the geometric ceiling, by phase (minute-30 entry throughout; green-day rate over both sides; 1,713 sessions).

Phase	Instrument	TP / SL (L)	TP / SL (ticks)	Ceiling $SL/(TP+SL)$	Realized	Unresolved
Evaluation	one full contract	0.64/0.96	128 / 192	60.0%	58.8% [57.1, 60.4]	2.6%
Funded lock	one full contract	1.20/0.96	240 / 192	44.4%	44.3% [42.6, 46.0]	5.4%
Fill (qualification)	three micros	0.15/0.30	100 / 200	66.7%	64.9% [63.3, 66.5]	2.1%

Table 3: The evaluation funnel (1,713 sessions; $N = 200{,}000$).

Quantity	Value
Green-day rate	58.8%
(green-day rate)^2	34.6%
P(pass), observed	34.73% [34.52, 34.94]
Expected value per evaluation, incl. withdrawals, at $\varphi = 0.05/0.07$	$+0.0716\,L / +0.0516\,L$ (± 0.0017)
Expected value per evaluation at list price, $\varphi = 0.14$	$-0.0184\,L$

5 Results

All estimates are on the 2020–2026 window unless stated otherwise. Bracketed ranges are 95% confidence intervals. Sample sizes, seeds and windows for every figure are in the appendix.

The geometry and the realized rates. The instrument differs by phase, so the two bracket columns are not proportional across rows: one tick is $0.005\,L$ on the full contract and $0.0015\,L$ on three micros. The qualification trade is sized in micros out of necessity rather than preference. On a full contract its $0.15\,L$ target would sit 30 ticks away, a distance the contract covers in seconds, and the agreement prohibits "disruptive practices … including but not limited to spoofing strategies, microscalping, hedging, and high frequency trading". Three micros spread the same target over 100 ticks, which the session takes time to travel. The instrument is therefore fixed by the product's rules, not chosen.

The evaluation funnel and the consistency rule. Passing requires two green days, so the pass rate obeys the $P \approx (\text{green-day rate})^2$ identity.
The base row never invokes a top-up day, which is the direct confirmation that the cap is met with equality. The last two rows apply a dispersion of ± 2 ticks to the realized amount of a target day; the $0.1\,L$ top-up is the size used in our own live template. The four rows of this table and of Table 5 are estimated at $N = 40{,}000$ so that they are comparable with one another; the converged estimate of expected value is the one reported in Table 3.
The sign of the cap is worth about $0.001\,L$ per evaluation to a participant who carries a top-up day sized to the shortfall, because a micro trade of a few ticks steps over the line, and about $0.049\,L$ to one who does not. The assumption is therefore not about the rule but about whether the participant has a cheap way to clear it.

The funded stage. After the first payout the floor stays locked but each withdrawal thins the cushion, so survival falls cycle after cycle.

Large-day-first yields about $0.025\,L$ more per account — roughly three standard errors, not noise — while large-day-last preserves accounts materially better: 65.4% death per cycle against 74.8%, and 17.2% of accounts reaching a second payout against 11.3%. Valuing a destroyed funded account at its replacement cost, the evaluation fee divided by the pass rate ($0.202\,L$), the preservation advantage is worth about $0.019\,L$ per cycle against a $0.025\,L$ shortfall in cash. The two schedules are within a few thousandths of L of each other and the ranking depends on

Table 4: Buyer expected value with and without the consistency rule ($N = 40{,}000$).

Evaluation target rule	P(pass)	EV at φ=0.05	EV at φ=0.07	Top-up days
Fixed target (no consistency rule)	34.8%	+0.073	+0.053	0.000
Moving cap, fixed bracket repeated — cap met with equality (base case)	34.8%	+0.073	+0.053	0.000
Moving cap, top-up sized to the shortfall	33.9%	+0.068	+0.048	0.268
Moving cap, top-up at a fixed $0.1\,L$	30.3%	+0.057	+0.037	0.270

Table 5: Sensitivity of the base case to the sign of the cap ($N = 40{,}000$; bracket repeated exactly).

Cap reading	Top-up day carried	P(pass)	EV at φ=0.05	EV at φ=0.07	Top-up days
Inclusive ($\leq \frac{1}{2}$) — base case	any or none	34.8%	+0.073	+0.053	0.000
Strict ($< \frac{1}{2}$)	sized to the shortfall	34.8%	+0.072	+0.052	0.338
Strict ($< \frac{1}{2}$)	fixed $0.1\,L$	29.9%	+0.056	+0.036	0.342
Strict ($< \frac{1}{2}$)	none — large days only	21.4%	+0.024	+0.004	0.000

whether account supply or cash per account is the binding constraint. The "0.2 × 5" schedule is worst on both counts.

The repeated purchase. The $K = 10$ row is the least precisely estimated and is reported at $N = 20{,}000$ rather than 3,000. Across twelve seeds at $N = 3{,}000$ the fifth percentile of profit ranges from $-1.955\,L$ to $+2.23\,L$, with a median of $+0.174\,L$, and is negative in five of the twelve; the net-loss probability ranges from 4.5% to 6.0%. The terminal fifth percentile at $K = 10$ is therefore not distinguishable from zero and we do not read a sign from it. The running minimum of the balance, reported below, is stable at $0\,L$ in all twelve seeds. The evaluations-purchased and fees columns are means and are carried from the $N = 3{,}000$ run, where they are stable.

Two stable comparisons show that the distribution is not a smooth ladder. First, the fraction of runs that end with no live accounts and no float is far larger than the probability of a net loss: at $K = 5$ and the fee paid, 3.4% of runs end below the initial stake but 25.5% end with the business dead, and at $K = 10$ the figures are 5.2% and 28.4%. The difference is runs that withdrew more than $2\,L$ before exhausting themselves. Second, the twenty-fifth percentile sits close to the median rather than midway to the lower tail: at $K = 10$ it is $+69.446\,L$ against a median of $+91.105\,L$, while the fifth percentile is at or below zero. Most of the probability mass is in a narrow upper band, and the rest is in a thin tail at the stake.

The float is a tail parameter rather than a return parameter: between $3\,L$ and $5\,L$ the median moves by 0.9% while the probability of a net loss falls from 1.3% to 0.1%.

We report the fifth percentile of the running minimum of the operating balance alongside the fifth percentile of the terminal outcome. They are different objects and they disagree completely: the terminal fifth percentile is positive at every K under both policies, while the balance itself passes through essentially zero — one to six thousandths of L — in the worst 5% of runs, at every K and under both policies. One quantity says how much the operator ends up with; the other says how close the operation comes, at its worst moment, to being unable to buy a single further evaluation. A venture whose median outcome is tens of L can spend part of its life in that state, and a table of terminal quantiles does not show it.

The buyer breaks even where the fee equals the expected cash a purchase returns, $\mathbb{E}[\text{cash}] = 0.122\,L$, against an observed φ of 0.07 and a list φ of 0.14. The number of evaluations bought is endogenous: it is whatever the float can finance before it is exhausted. At $\varphi = 0.05$ a run buys 422 evaluations and only 4% of runs end with no accounts and no float; at list price it

Table 6: Locking the floor in one day against two ($N = 20{,}000$).

Approach	P(reaches a payout)	Expected cash (L)
One day	35.4%	0.36
Two days	30.7%	0.296

Table 7: Withdrawal schedules over the life of a funded account from a cushion of $0.6\,L$ ($N = 60{,}000$).

Schedule	Life (L/account)	Death per cycle	Reaches a second payout
Large day last (four minimum days, then push)	+0.427 (±0.005)	65.4%	17.2%
Large day first	+0.452 (±0.006)	74.8%	11.3%
"0.2 × 5"	+0.215 (±0.002)	74.3%	9.5%

buys 89 and 98.6% of runs end exhausted. This is why a negative expected value per evaluation ($-0.0184\,L$) is still consistent with a positive median outcome ($+0.792\,L$): the loss is truncated at the $2\,L$ stake while the gains are not, and at list price the venture does not get enough trials for the negative drift to compound. The median is not a measure of edge here; it is a measure of how few bets the buyer can afford before the door closes. The gap between promotional and list pricing is therefore the whole of the buyer's expected value: at list price it is negative.

A rule-structure artifact. We note a structural artifact of the rule set. Two accounts on opposite sides of the same session win at least one 99.8% of the time, against 83.0% if the two outcomes were independent — 16.8 percentage points extracted from the rule structure rather than from the market (funded lock: 88.6% against 69.0%, +19.6 points; fill: 99.8% against 87.7%, +12.1 points). We report this as a property of the design, not as a usable strategy. The agreements we examined prohibit not only "simultaneously entering into opposite positions" between connected accounts but also, in the same clause, any activity "meant to pool or hedge risk of the aggregate, unconnected accounts", and operating several accounts as a single book against a shared bankroll falls within that language on its face. The multi-account figures above should therefore be read as an upper bound on what the rule structure would concede if it were permitted, not as a design a buyer can implement.

The seller's side. A funded account is simulated, so the operator's only cash outflow is the payout. Writing $V = \mathbb{E}[\text{payout} \mid \text{funded}]$ for the expected total payout of a funded account and φ for the fee, the operator breaks even against a population whose pass rate is $p^* = \varphi/(V/L)$. On our estimates $V/L = 0.35$, so $p^* = \varphi/0.35$: 14.3% at $\varphi = 0.05$, 20.0% at 0.07, 28.6% at 0.10 and 40.0% at 0.14. The mechanical participant we model passes at 34.7%. The two facts are consistent only if the population passes at well under two-thirds of the mechanical rate — which is the mechanism the supervisory warnings describe.

The estimate is conservative in the relevant direction. V is computed for our disciplined participant; a less disciplined population reaches fewer payouts per funded account, so its V is lower and p^* correspondingly higher. At half our V the break-even rate at $\varphi = 0.07$ is 40.0% — above the 34.7% attained by the best configuration anywhere in our sizing sweep — so the operator would be profitable even against the optimised buyer. Our figure is thus the most demanding version of the argument, not the most convenient one.

Note the symmetry with Table 12. At list price the operator is profitable against any buyer whose pass rate is below 40%, which includes the best rate we attain, while the buyer's own expected value per evaluation has already turned negative. List price is where the product stops being a bet with a contested edge and becomes a sale. The buyer's positive expectation

Table 8: Outcome by book size (K accounts; simultaneous-entry policy; float $2\,L$, each payout withdrawn; 704-session horizon; $\varphi = 0.07$; $N = 3{,}000$, except the $K = 10$ row ($N = 20{,}000$; see below)).

K	Median profit (L)	P(net loss)	5th pct (L)	Evaluations	Fees (L)	Runs exhausted
3	+27.946	3.5%	+1.79	230	16.118	21.6%
5	+46.286	3.4%	+3.075	370	25.885	25.5%
8	+73.302	3.6%	+4.428	584	40.868	27.1%
10	+91.105	5.2%	−0.882	709	49.621	28.4%

Table 9: Panel B — sensitivity to a cap on accounts purchased over the horizon ($K = 5$, $\varphi = 0.07$).

Cap	Median profit (L)	P(net loss)	5th pct profit (L)	Evaluations used
10	+0.755	31.9%	−0.7	10
20	+1.891	18.6%	−0.917	20
50	+5.175	6.4%	−0.347	50
100	+11.19	4.0%	+1.485	97
none (as reported)	+46.286	3.4%	+3.075	370

at promotional prices and the seller's margin are not in contradiction: they are the two tails of one distribution, and the fee is the parameter that selects which tail the operator is selling into.

6 Discussion

The product is marketed on trading skill. The economics are computed for a participant with no directional edge. Any real edge moves them upward, so the reported values are a lower bound on what a skilled buyer obtains and, correspondingly, an upper bound on what the seller can rely on. We note the limits of our instrument rather than claim to have settled the question: on the estimation window the evaluation bracket requires a drift of about 34 index points per session — roughly 0.15% of the underlying — to register a two-standard-error effect on the per-day win rate, so a small conditional edge would not be visible in it. A session-level test of the entry levels against a matched random-side control gives $-0.005R$ with a 95% interval of $[-0.026, +0.016]$: the level supplies no direction that we can detect, at a minimum detectable effect of $0.021R$. Nothing in this paper turns on whether a directional edge exists.

What the model does establish is that the geometry sets a ceiling and the design determines how far below it the participant lands. The bare identity gives 44.4%. The daily bracket and the truncation of its stop bring that to 40.4%. The trailing floor brings it to 34.7%. None of the three is visible from the marketing.

The headline flatters the product — a large notional funded account and a generous profit split — while the terms that set its value sit in the fine print. Not every term that reads as a hurdle bites: the moving consistency target is met with equality by a repeated fixed bracket. The levers that do bite are structural. Until the floor locks, winning does not widen the cushion, so the single lock day is the bottleneck. After the first payout every withdrawal thins the cushion, so most funded accounts die not on the lock day but in the routine days of later cycles.

Netted out, the expected value per evaluation is modestly positive at the fee actually paid and negative at list price. Because the accounts are simulated and the provider is the counterparty, a payout is a marketing and retention expense rather than a share of market proceeds — an arrangement whose incentives favour churn. Persistent demand for a roughly break-even-to-negative product is consistent with a taste for lottery-like bets, a fixed and known stake for

Table 10: Panel C — sensitivity to the size of the float ($K = 10$, $\varphi = 0.07$; $N = 20{,}000$, same seed as the $K = 10$ row of Table 8, with which the $2\,L$ row coincides).

Float (L)	Median profit (L)	P(net loss)	5th pct profit (L)	Balance minimum (L)
1	+32.887	21.4%	−0.98	+0.001
2	+91.105	5.2%	−0.882	+0
3	+95.224	1.3%	+54.524	+0.009
5	+96.117	0.1%	+71.604	+0.48

Table 11: Panel D — the running minimum of the operating balance ($\varphi = 0.07$, float $2\,L$, $N = 3{,}000$).

K	5th pct profit (simultaneous)	5th pct of the balance minimum	5th pct profit (queue)	5th pct of the balance minimum (queue)
3	+2.175	+0.005	+4.209	+0.006
5	+2.247	+0.002	+8.977	+0.005
8	+3.346	+0.001	+13.816	+0.002
10	−0.882	+0	+22.448	+0.002

a small chance at a large and salient payout (Kumar, 2009; Barberis, 2012), and with the overweighting of small-probability, large-gain outcomes familiar from cumulative prospect theory (Tversky and Kahneman, 1992).

7 Scope, limitations, and conclusion

Scope. We restrict attention to the rule geometry of the product and to a participant with no directional edge. The economics are a lower bound on what a buyer with an edge obtains.

Limitations. Five bear on how far these figures travel.

First, the multi-account figures may not describe a permitted activity. The agreement we examined prohibits, in a single clause, trades "the purpose of which is to manipulate, abuse, or give you an unfair advantage…by…simultaneously entering into opposite positions", and separately any activity, including between unconnected accounts, "meant to pool or hedge risk of the aggregate, unconnected accounts". Running K accounts as one book against a shared bankroll falls within the second phrase on its face, whether or not positions overlap in time. Tables 8 and 12 should be read as a characterisation of what the rule structure would concede, not as an operating design.

Second, the discretionary levers are not modelled and are not diversifiable. The agreement reserves termination "at any time for any reason or no reason", subjects "excessive or unrealistic single day or single trade profits" to review, and on a determination of prohibited conduct forfeits accrued rewards and requires the return of rewards already received. The single lock day we price is by construction a deliberate 4.8% single-trade gain, which is the conduct that clause names. A per-payout denial probability is easy to accommodate — at $q = 0.10$ the median at $K = 5$ falls from $46.286\,L$ to $37.226\,L$ — but a clawback is a single correlated event that removes the venture's entire realised return, and we do not attempt to price it.

Third, counterparty risk is not modelled at all. The accounts are simulated and the balances are not client money.

Fourth, we do not know whether a cap exists on accounts or allocated capital per customer. The tables assume none, and the assumption does the work: the $46.286\,L$ median at $K = 5$

Table 12: Outcome against entry fee ($K = 5$, float $2\,L$).

Entry fee φ	Median profit (L)	P(net loss)	Evaluations	Runs exhausted	EV per evaluation (L)
0.05 (promotional)	+56.551	0.6%	422	4.1%	+0.0716
0.07 (as paid)	+46.286	3.4%	370	25.5%	+0.0516
0.1	+14.782	17.7%	218	77.7%	+0.0216
0.14 (list)	+0.792	45.5%	89	98.6%	−0.0184

requires buying 370 evaluations from one provider over three trading years. Capping the horizon at 50 accounts — still an order of magnitude above ordinary use — reduces the median to $5.175\,L$ and raises the net-loss probability from 3.4% to 6.4%.

Fifth, neither the data nor the simulation code accompany this paper, so the reported figures are not independently verifiable as published. We describe the barrier logic, the account parameters, the death convention and the estimation windows in enough detail for the design to be re-implemented, and we report the simulation seeds and sample sizes for every table in the appendix, but re-implementation is not replication.

Two further limitations are stated where they arise: the evaluation-stage rules are known only from a live dashboard on two accounts (Section 2), and the sign of the consistency cap and the rounding of the displayed target are not established by any document we hold (Section 3).

Conclusion. Stripped to its core, an evaluation challenge is a barrier bet whose probability of success is bounded by the loss limit over the sum of loss limit and profit target, invariant to the market's volatility. The bound is not attained: the trailing floor, the discretisation of the daily bracket and the friction charged per trade each move the realized rate below it, and the ceiling is unreachable from either direction of position size. Layered on top, the design levers determine the buyer's expected value, and the entry price determines its sign. Under the pricing actually observed, and abstracting from the constraints set out above, the arithmetic of the product favours the repeat buyer; at list price it does not.

Disclosure

The author holds and has traded funded accounts with providers of the type analyzed here and has received payouts from them during the sample period. The author also produces educational material and commercial trading indicators for retail traders. No provider is named, no provider funded or reviewed this research, and the analysis reflects only the author's independent work.

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A Estimation details for every reported figure

Death convention, used throughout. The daily stop is truncated to the cushion, $sl = \min(sl_{\text{nominal}}, \text{cushion} - 0.04\,L)$; the account dies when $sl < 0.04\,L$; and the race is recomputed with the truncated stop, so a truncated stop also lowers the probability of reaching the target. Barriers are evaluated on the full intrabar path. The modelled participant abandons an evaluation after 45 trading days, a parameter of the participant rather than of the product.

Two estimation windows. The economics, the power calculation and the position-size sweep are estimated on 1,713 sessions spanning 2020–2026. The directional tests are estimated on a 704-session sub-window (2023–2026), the only span for which the pre-registered grid's session features are constructed. The "704" appearing in the horizon of Tables 8 and 12 is a number of simulated days forward, not a data window: paths are drawn with replacement from the 1,713 sessions.

Unit. Every monetary value below is in L.

Figure	Window (sessions)	N	Seed	Value
Table 2, Evaluation	1,713	census	—	58.8% [57.1, 60.4]; unresolved 2.6%
Table 2, Funded lock	1,713	census	—	44.3% [42.6, 46.0]; unresolved 5.4%
Table 2, Fill	1,713	census	—	64.9% [63.3, 66.5]; unresolved 2.1%
Table 3, green-day rate	1,713	census	—	58.8%
Table 3, $(\text{rate})^2$	analytic	—	—	34.6%
Table 3, P(pass)	1,713	200,000	4242	34.73% [34.52, 34.94]
Table 3, EV at 0.05 / 0.07 / 0.14	1,713	200,000	4242	$+0.0716/+0.0516\ (\pm 0.0017)/-0.0184$
Table 4, four rows	1,713	40,000	4242	34.8/34.8/33.9/30.3%
Section 3, margin over the effective target	704 (contiguous accounts)	564 accounts	$4242+i$	97% of two-day passes under 0.001 of margin; none under the alternative design
Table 5, four rows	1,713	40,000	4242	34.8/34.8/29.9/21.4%
Table 6	1,713	20,000	4242	35.4%/0.36; 30.7%/0.296
Table 7, three rows	1,713	60,000	7 (life) / 2026 (cycle)	$+0.427\ (\pm 0.005)/+0.452\ (\pm 0.006)/+0.215\ (\pm 0.002)$
Table 8, medians and P(net loss), $K = 3, 5, 8$	1,713; horizon 704 days	3,000	$4242+K+$fee	$+27.946/+46.286/+73.302$

Figure	Window (sessions)	N	Seed	Value
Table 8, $K = 10$ row (median, P(net loss), 5th and 25th pct, exhaustion, balance minimum)	1,713; horizon 704 days	20,000	4322	+91.105; 5.2%; −0.882; +69.446; 28.4%; +0
Table 8, evaluations purchased	as above	3,000	as above	230/370/584/709
Table 8, runs exhausted, $K = 3, 5, 8$	as above	3,000	as above	21.6/25.5/27.1%
Table 8, $K = 10$ stability	as above	3,000 × 12 seeds	12 seeds	5th pct −1.955 to +2.23, negative in 5 of 12; P(net loss) 4.5–6.0%
Table 9 (account cap)	as above	3,000	as above	$+0.755/+1.891/+5.175/+11.19$
Table 10 (float)	as above	20,000	4322	21.4/5.2/1.3/0.1%
Table 11 (balance minimum)	as above	3,000, except $K = 10$ (20,000)	4242+K+fee + float	$+0.005/+0.002/+0.001/+0$
Table 11, queue policy	as above, 1-minute entry grid	3,000	as above	+26.427/ + 40.727/ + 57.592/ + 65.721
Table 12, four rows	as above	3,000	as above	0.6/3.4/17.7/45.5%
Table 12, evaluations and exhaustion	as above	3,000	as above	422/370/218/89; 4.1/25.5/77.7/98.6%
Section 3, ceiling	analytic	—	—	$L/(T + L) = 1/2.25 = 44.4\%$
Section 3, friction drift	1,713	census	4242	−0.0191 per session (−1.9% of the cushion); −1.7 points
Section 3, lock ceiling	analytic	—	—	$192/(240 + 192) = 44.4\%$
Section 3, size sweep	1,713	50,000	4242	40.4% fixed floor; 34.7% trailing; 28.4% and 0.5% at smaller sizes
Section 4, unresolved by minute	1,713	census	4242	1.9% at minute 0; 8.0% at minute 180
Section 4, power of the bracket	1,713	census	4242	34 index points per session (2σ = 2.37 points of win rate)
Section 4, directional test	704	B = 20,000	11	$-0.005R$ $[-0.026, +0.016]$; minimum detectable effect $0.021R$
Section 4, account policies	1,713; horizon 704 days	3,000	4242+K+fee	44.723 / 2.0%; 46.286 / 3.4%; 2.82 / 39.7%
Section 5, rule-structure artifact	1,713	census	—	99.8 vs 83.0; 88.6 vs 69.0; 99.8 vs 87.7
Section 5, seller break-even	1,713	200,000	4242	$V = 0.3501$; $p^* = 14.3/20.0/28.6/40.0\%$
Section 7, payout-denial sensitivity	1,713; horizon 704 days	3,000	4242+K+fee	median 46.286 → 37.226 at $q = 0.10$